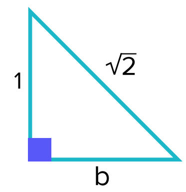
Special right triangles

1 In a $30^\circ - 60^\circ - 90^\circ$ triangle, identify which leg is shorter.

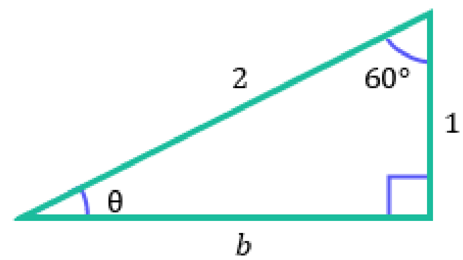
2 Find the exact length of b for the given triangle.



3 Consider the following triangle.

a Find the exact value of b.

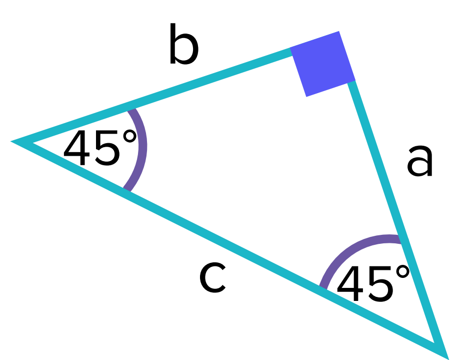
b Find the value of θ.



4 Consider the triangle as shown.

a Complete the table.

a	1	2	3	4	5
b					
c					



b Let a be x units. Find b in terms of x.

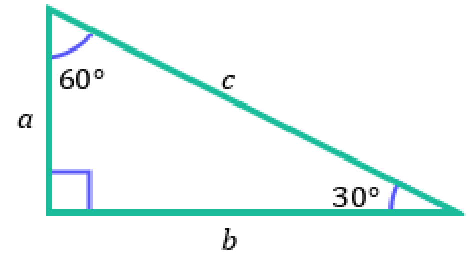
c Find c in terms of x.

d Describe the relationship between the sides of any isosceles right triangle.

5 Consider the triangle as shown.

a Complete the table.

a	1	2	3	4	5
b					
c	2	4	6	8	10



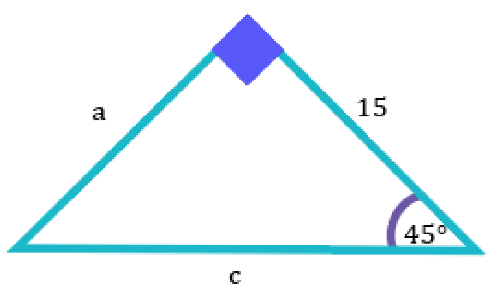
b Let a be x units and c is $2x$ units. Find b in terms of x.

6 For each of the following triangles:

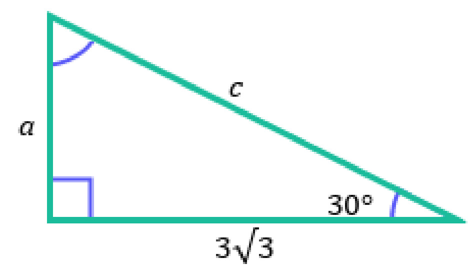
i Find the exact value of a.

ii Find the exact value of c.

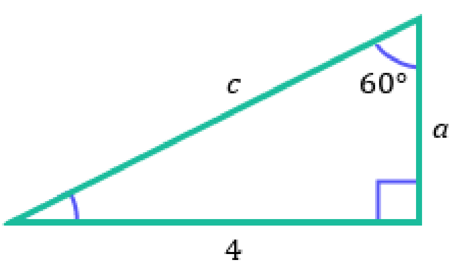
a



b



c

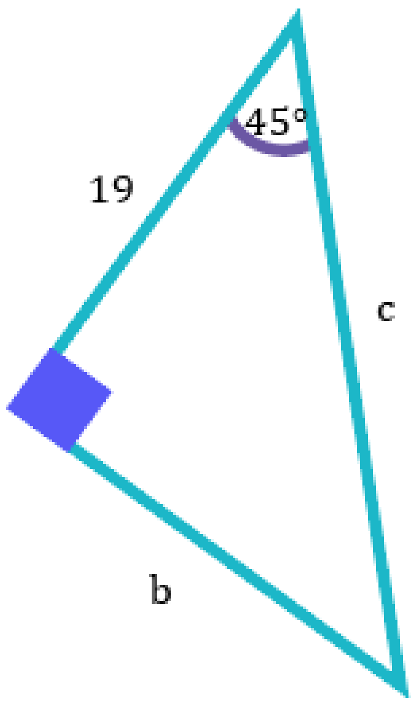


7 For each of the following triangles:

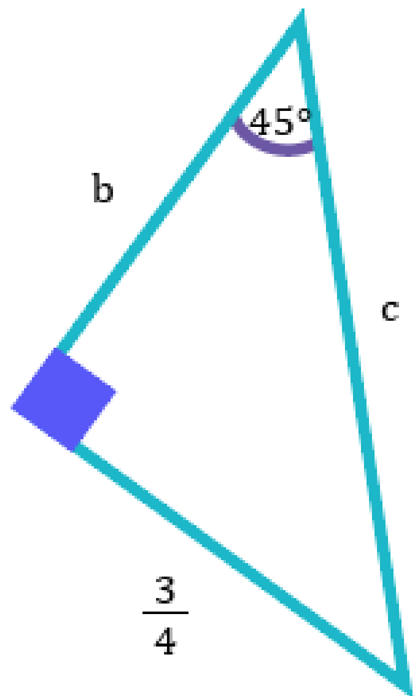
i Find the exact value of b.

ii Find the exact value of c.

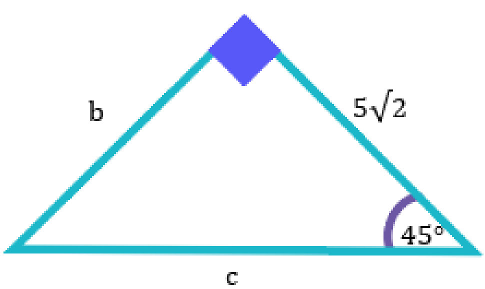
a



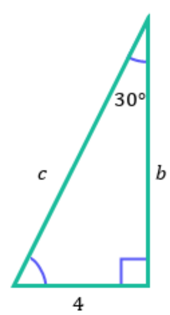
b



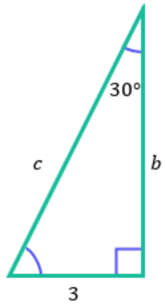
c

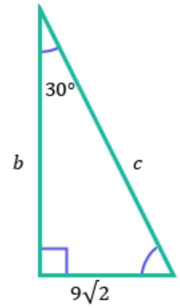


d



e



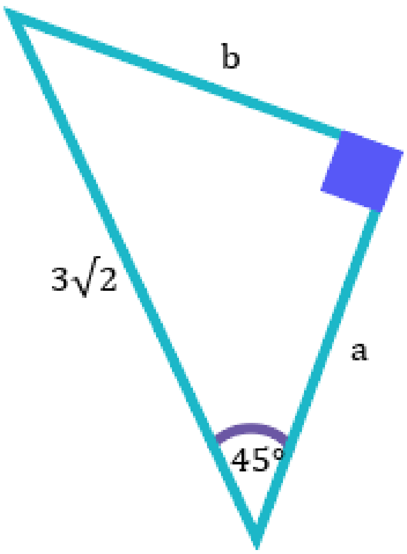


8 For each of the following triangles:

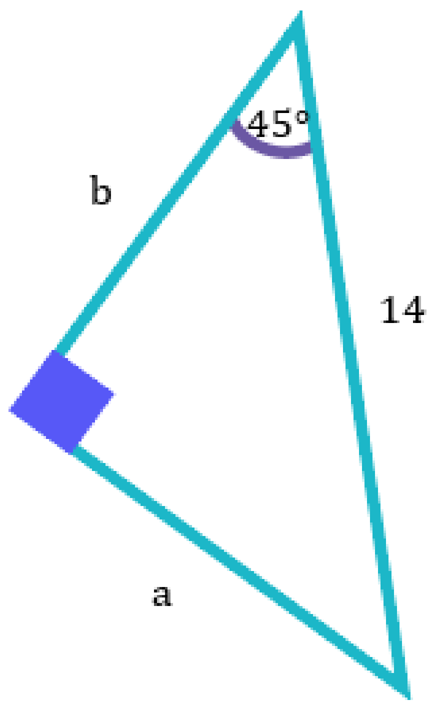
i Find the exact value of a.

ii Find the exact value of b.

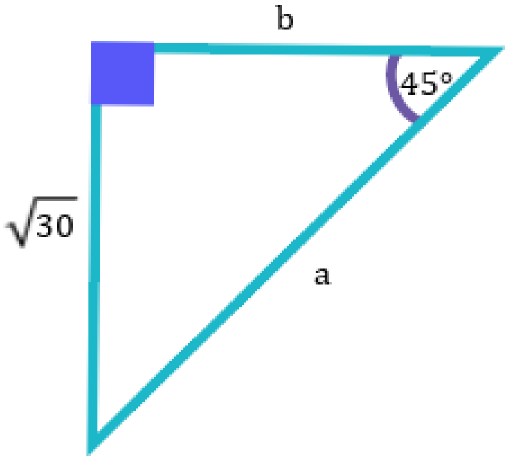
a

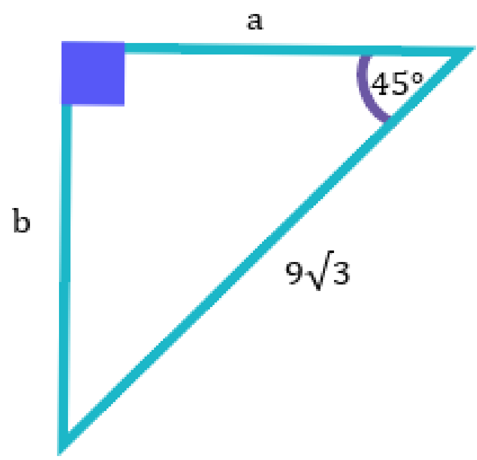


b



c





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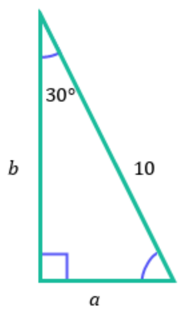

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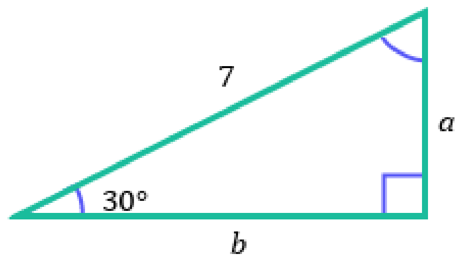
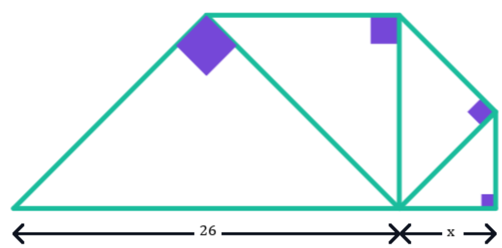
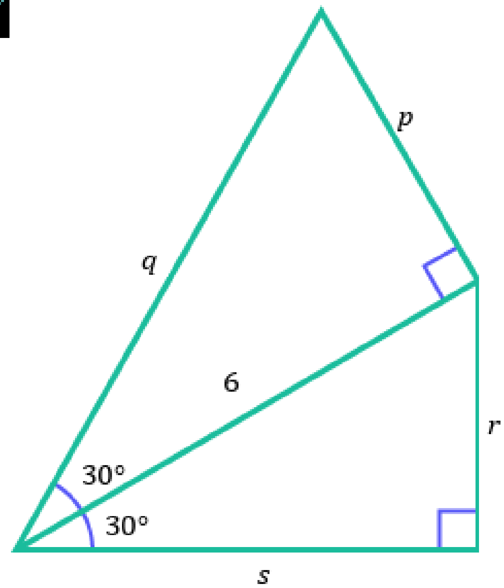


9 Each triangle in the following diagram is right-angled and isosceles.
Find the exact value of x.



10 Consider the following diagram.



a Find the exact length of p.

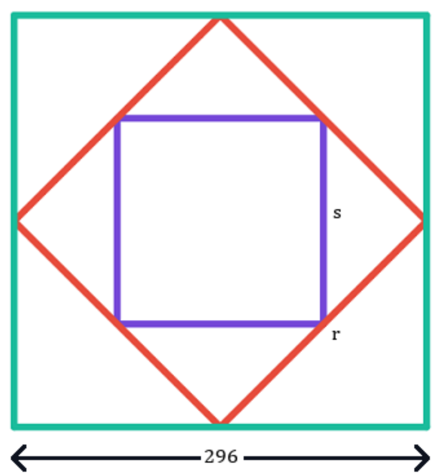
b Find the exact length of q.

c Find the exact length of r.

d Find the exact length of s.

e Hence, find the perimeter of the quadrilateral.

11 A computer graphic is created by first connecting the midpoints of a square that is 296 pixels by 296 pixels to form a smaller square within it and then by repeating first step on this smaller square to get an even smaller one.



a Find the exact length of r.

b Hence, find the exact length of s.

12 The diagonal of a square measures 15 units.

a Find the exact side length of the square.

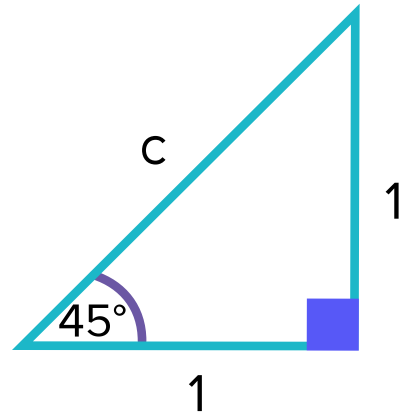
b Hence, find the perimeter of the square.

13 Oprah is building a table in the shape of an equilateral triangle. She only has $8\ \mathrm{m}$ of trim wood to go around the perimeter of the table.
Find the exact maximum area for the surface of the table that she can build.

14 The area of an equilateral triangle is $36\sqrt{3}\ \mathrm{cm}^2$.
Find the perimeter of the triangle.

Additional questions

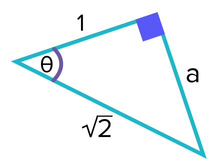
15 Find the exact length of c.



16 For the given triangle:

a Find the value of a.

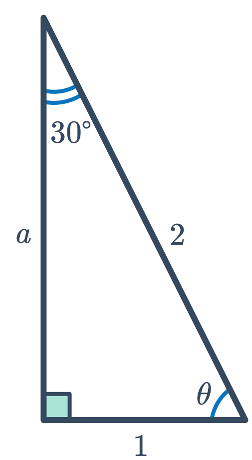
b Find the value of θ.



17 For the given triangle:

a Find the exact value of a.

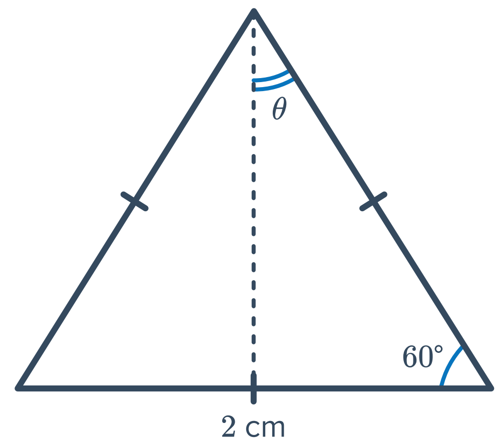
b Find the value of θ.



18 The equilateral triangle has side lengths of [redacted] The perpendicular height of the triangle is shown in the diagram.

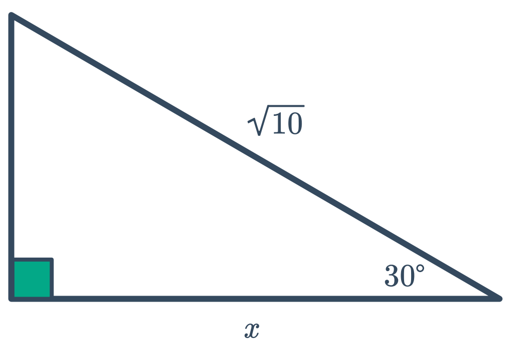
a Find θ.

b Find the exact value of the perimeter of one of the smaller triangles.

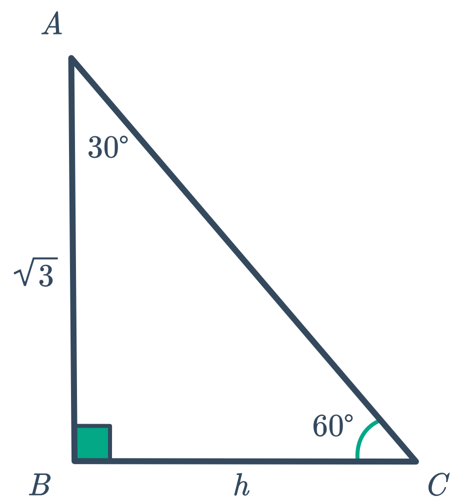


19 Find the value of the variable(s) in the following triangles. Express your answer in simplest radical form.

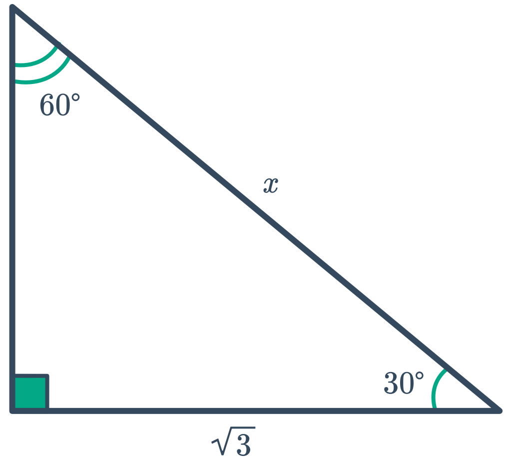
a

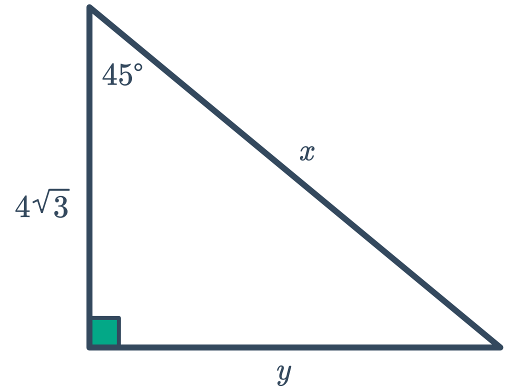


b



c





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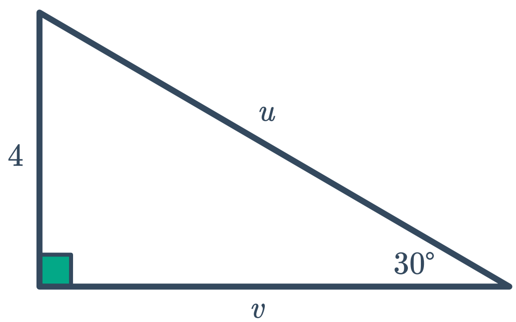

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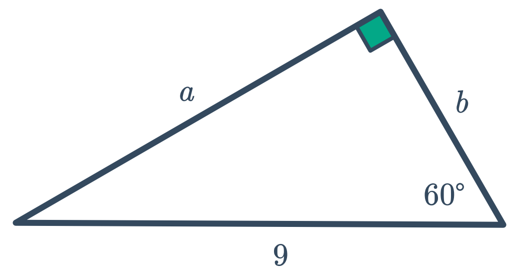

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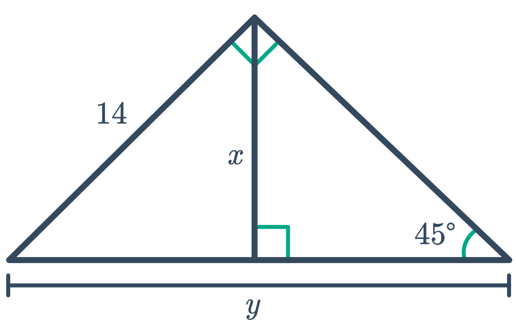

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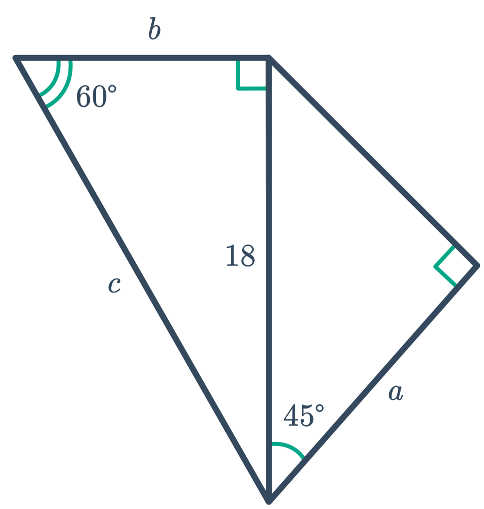
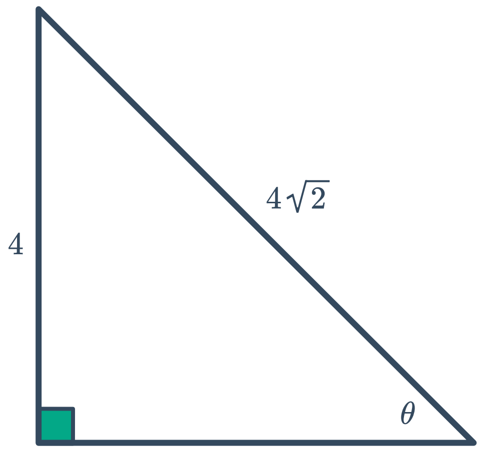
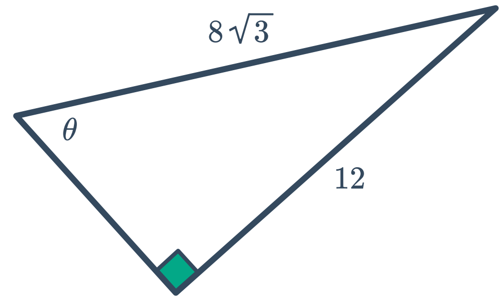


20 Find θ in the following triangles once the si[redacted]ngths are known:

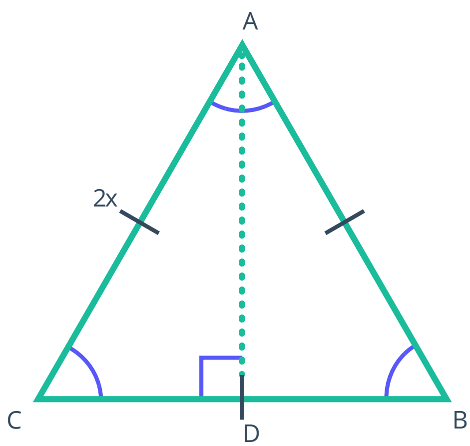
a



b



21 $\triangle ABC$ is an equilateral triangle with side lengths of $2x$ units. An altitude is drawn from A to meet side $\overline{BC}$ at D. This information is shown in the diagram below.



a Find the $m\angle B$.

b Given that $\overline{AD}$ bisects $\angle BAC$, find the measure of $\angle BAD$.

c Given that $\overline{AD}$ bisects $\overline{BC}$, find the length of $\overline{BD}$.

d Determine the length of the altitude $\overline{AD}$ in terms of x.

e State whether the following statements [redacted]ue or false. If false, correct the statement so it is true.

i The longer leg of a $30^\circ - 60^\circ - 90^\circ$ triangle is twice the length of the shorter leg.

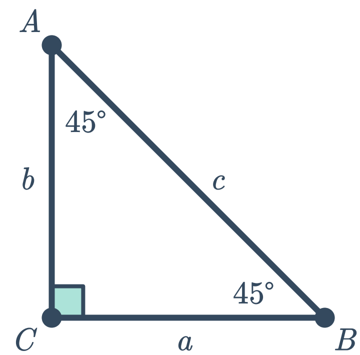
ii The shorter leg of a 30°-60°-90° triangle is opposite the angle measuring 30°.

iii The hypotenuse of a $30^\circ - 60^\circ - 90^\circ$ triangle is $\sqrt{3}$ times the length of the longer leg.

iv The ratio of sides in a $30^\circ - 60^\circ - 90^\circ$ triangle is $1 : \sqrt{3} : 2$.

22 Find the exact side length of an equilateral triangle with a perpendicular height of $\sqrt{21}$ in.

23 Given: $\triangle ABC$ is a $45^\circ - 45^\circ - 90^\circ$ triangle
Prove: $c = a\sqrt{2}$



24 Zeki claims that all isosceles right triangles are similar to each other. Determine if Zeki is correct. Explain your reasoning.

25 Vinod is trying to build this bookcase, but does not know all of the interior angles of the triangle. Vinod knows that the hypotenuse is $8\,\text{ft}$ and the base is $4\,\text{ft}$. Explain why a special right triangle can be used to solve the problem.